

Rubric for Assessment of Proposals for Group Design ProjectTechnical Content:

Executive summary on cover page:

- Summarized goals
- Identified project location
- Stated power demand to be satisfied

Motivation for project addressed

Power demand:

- Identified/estimated
- Identified/estimated based on scale/location
- How identified/estimated discussed

Constraints on development:

- Economic constraints identified/discussed
- Environmental constraints identified/discussed
- Political constraints identified/discussed
- Health/safety constraints identified/discussed
- Reliability/sustainability constraints identified/discussed

Additional questions/items to be considered addressed:

Advantages/disadvantages of conventional technology:

- Advantages identified (minimum of 2)
- Disadvantages identified (minimum of 2)
- Role of scale/location considered

Background on sustainable/renewable technology:

- History of technology summarized
- 2-5 significant developments identified
- Current state-of-the-art identified/discussed
- Role of scale/location considered

Advantages/disadvantages of sustainable/renewable technology:

- Advantages identified (minimum of 2)
- Disadvantages (minimum of 2)
- Role of scale/location considered

Preliminary specifications for sustainable/renewable technology:

- Specific specifications for technology identified
- Source for specifications/process for estimating specifications noted

Advantages/disadvantages of air/liquid cooling technologies:

- Advantages identified (minimum of 2)
- Disadvantages (minimum of 2)
- Role of scale/location considered

Background on air/liquid cooling technology:

- History of technology summarized

- 2-5 significant developments identified

- Current state-of-the-art identified/discussed

- Role of scale/location considered

Preliminary specifications for air/liquid cooling technology:

- Specific specifications for technology identified

- Source for specifications/process for estimating specifications noted

Power savings approaches identified

Quality of Writing:

Submission:

- No more than 6 pages (not including cover page, references, & appendices)
- Single-spaced
- 1" margins throughout

Cover page:

- Included project name & date
- Included names of each team member
- Summary paragraph was concise

Appendices:

- Used appropriately
- Organized to guide reader

References:

- Included
- Formatted correctly

Writing:

- No errors in spelling
- No errors in grammar
- Clarity
- Conciseness
- Professionalism of writing (*i.e.*, use of language & subheadings)

Assessment: Will receive a score of Meets Expectations if there are moderate deficiencies in no more than 2 sections of technical content and the writing is largely free of typos/errors, easy to follow/understand, and includes all required sections (bibliography, citations in text, appendices, cover page, etc.). Scores of Excellent can be achieved for reports that contain minor deficiencies in no more than 2 sections of technical content and the writing and formatting is of a highly professional and polished quality.

A minor deficiency for example might be a math/calculator error or a small misunderstanding of the material. A moderate deficiency for example might be a section that is missing, did not include all the noted information, or contains more significant math errors involving mishandling of units, etc.